```
❯ gcc -o run challenge.c -lm && ./run
Iter  0 [ 28x  7, density=0.19, nnz=  32]: PASS (Max error: 0.00e+00)
Iter  1 [ 37x 26, density=0.36, nnz= 348]: PASS (Max error: 0.00e+00)
Iter  2 [ 37x 29, density=0.23, nnz= 254]: PASS (Max error: 0.00e+00)
Iter  3 [ 23x 44, density=0.27, nnz= 282]: PASS (Max error: 0.00e+00)
Iter  4 [ 36x 31, density=0.09, nnz= 103]: PASS (Max error: 0.00e+00)
Iter  5 [  5x 24, density=0.18, nnz=  24]: PASS (Max error: 0.00e+00)
Iter  6 [ 45x  8, density=0.14, nnz=  48]: PASS (Max error: 0.00e+00)
Iter  7 [ 29x 22, density=0.10, nnz=  67]: PASS (Max error: 0.00e+00)
Iter  8 [  9x 24, density=0.11, nnz=  27]: PASS (Max error: 0.00e+00)
Iter  9 [ 35x 39, density=0.37, nnz= 530]: PASS (Max error: 0.00e+00)
Iter 10 [ 14x 44, density=0.27, nnz= 157]: PASS (Max error: 0.00e+00)
Iter 11 [ 34x  9, density=0.25, nnz=  78]: PASS (Max error: 0.00e+00)
Iter 12 [  8x 38, density=0.19, nnz=  68]: PASS (Max error: 0.00e+00)
Iter 13 [ 43x 45, density=0.35, nnz= 644]: PASS (Max error: 0.00e+00)
Iter 14 [  8x 12, density=0.15, nnz=  18]: PASS (Max error: 0.00e+00)
Iter 15 [ 40x  9, density=0.13, nnz=  50]: PASS (Max error: 0.00e+00)
Iter 16 [ 44x 38, density=0.10, nnz= 201]: PASS (Max error: 0.00e+00)
Iter 17 [ 45x 36, density=0.39, nnz= 611]: PASS (Max error: 0.00e+00)
Iter 18 [  9x 45, density=0.17, nnz=  60]: PASS (Max error: 0.00e+00)
Iter 19 [ 23x 12, density=0.24, nnz=  62]: PASS (Max error: 0.00e+00)
Iter 20 [ 26x  5, density=0.27, nnz=  30]: PASS (Max error: 0.00e+00)
Iter 21 [ 30x 14, density=0.05, nnz=  26]: PASS (Max error: 0.00e+00)
Iter 22 [  5x 12, density=0.16, nnz=  11]: PASS (Max error: 0.00e+00)
Iter 23 [ 38x 27, density=0.25, nnz= 243]: PASS (Max error: 0.00e+00)
Iter 24 [ 29x 32, density=0.15, nnz= 150]: PASS (Max error: 0.00e+00)
Iter 25 [ 35x 17, density=0.09, nnz=  53]: PASS (Max error: 0.00e+00)
Iter 26 [ 24x 32, density=0.24, nnz= 200]: PASS (Max error: 0.00e+00)
Iter 27 [ 36x 18, density=0.37, nnz= 247]: PASS (Max error: 0.00e+00)
Iter 28 [ 13x 45, density=0.10, nnz=  61]: PASS (Max error: 0.00e+00)
Iter 29 [  6x 27, density=0.08, nnz=  12]: PASS (Max error: 0.00e+00)
Iter 30 [ 24x 21, density=0.28, nnz= 117]: PASS (Max error: 0.00e+00)
Iter 31 [ 30x 19, density=0.05, nnz=  25]: PASS (Max error: 0.00e+00)
Iter 32 [ 23x 25, density=0.38, nnz= 208]: PASS (Max error: 0.00e+00)
Iter 33 [ 11x 41, density=0.33, nnz= 154]: PASS (Max error: 0.00e+00)
Iter 34 [ 11x 41, density=0.33, nnz= 146]: PASS (Max error: 0.00e+00)
Iter 35 [ 17x  5, density=0.21, nnz=  22]: PASS (Max error: 0.00e+00)
Iter 36 [ 44x 31, density=0.36, nnz= 499]: PASS (Max error: 0.00e+00)
Iter 37 [ 44x 28, density=0.10, nnz= 142]: PASS (Max error: 0.00e+00)
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Iter 38 [ 17x 14, density=0.20, nnz=  47]: PASS (Max error: 0.00e+00)
Iter 39 [ 40x 37, density=0.29, nnz= 435]: PASS (Max error: 0.00e+00)
Iter 40 [  7x 37, density=0.29, nnz=  79]: PASS (Max error: 0.00e+00)
Iter 41 [ 32x 17, density=0.15, nnz=  93]: PASS (Max error: 0.00e+00)
Iter 42 [  9x 26, density=0.34, nnz=  89]: PASS (Max error: 0.00e+00)
Iter 43 [ 11x 17, density=0.37, nnz=  64]: PASS (Max error: 0.00e+00)
Iter 44 [ 30x  7, density=0.36, nnz=  85]: PASS (Max error: 0.00e+00)
Iter 45 [ 26x 18, density=0.10, nnz=  45]: PASS (Max error: 0.00e+00)
Iter 46 [ 25x 45, density=0.16, nnz= 167]: PASS (Max error: 0.00e+00)
Iter 47 [  9x  5, density=0.14, nnz=   4]: PASS (Max error: 0.00e+00)
Iter 48 [ 23x  8, density=0.15, nnz=  26]: PASS (Max error: 0.00e+00)
Iter 49 [ 24x  9, density=0.22, nnz=  49]: PASS (Max error: 0.00e+00)
Iter 50 [ 15x 25, density=0.12, nnz=  48]: PASS (Max error: 0.00e+00)
Iter 51 [ 24x 11, density=0.16, nnz=  47]: PASS (Max error: 0.00e+00)
Iter 52 [  6x  8, density=0.09, nnz=   9]: PASS (Max error: 0.00e+00)
Iter 53 [ 19x  7, density=0.29, nnz=  32]: PASS (Max error: 0.00e+00)
Iter 54 [ 42x 39, density=0.23, nnz= 361]: PASS (Max error: 0.00e+00)
Iter 55 [ 32x  7, density=0.17, nnz=  35]: PASS (Max error: 0.00e+00)
Iter 56 [ 34x 35, density=0.15, nnz= 185]: PASS (Max error: 0.00e+00)
Iter 57 [ 43x 30, density=0.33, nnz= 425]: PASS (Max error: 0.00e+00)
Iter 58 [ 15x 33, density=0.29, nnz= 134]: PASS (Max error: 0.00e+00)
Iter 59 [ 15x 18, density=0.22, nnz=  62]: PASS (Max error: 0.00e+00)
Iter 60 [  6x 31, density=0.27, nnz=  48]: PASS (Max error: 0.00e+00)
Iter 61 [ 41x 26, density=0.14, nnz= 140]: PASS (Max error: 0.00e+00)
Iter 62 [ 36x 14, density=0.21, nnz= 126]: PASS (Max error: 0.00e+00)
Iter 63 [ 28x 24, density=0.11, nnz=  67]: PASS (Max error: 0.00e+00)
Iter 64 [ 15x 23, density=0.31, nnz= 117]: PASS (Max error: 0.00e+00)
Iter 65 [ 38x 25, density=0.35, nnz= 325]: PASS (Max error: 0.00e+00)
Iter 66 [ 25x 44, density=0.25, nnz= 268]: PASS (Max error: 0.00e+00)
Iter 67 [ 28x  5, density=0.32, nnz=  43]: PASS (Max error: 0.00e+00)
Iter 68 [ 14x 26, density=0.21, nnz=  74]: PASS (Max error: 0.00e+00)
Iter 69 [ 30x 31, density=0.10, nnz=  93]: PASS (Max error: 0.00e+00)
Iter 70 [ 16x  9, density=0.08, nnz=  13]: PASS (Max error: 0.00e+00)
Iter 71 [ 13x  8, density=0.35, nnz=  39]: PASS (Max error: 0.00e+00)
Iter 72 [ 18x 13, density=0.22, nnz=  46]: PASS (Max error: 0.00e+00)
Iter 73 [ 21x 38, density=0.31, nnz= 251]: PASS (Max error: 0.00e+00)
Iter 74 [  6x 29, density=0.19, nnz=  42]: PASS (Max error: 0.00e+00)
Iter 75 [ 40x 37, density=0.34, nnz= 476]: PASS (Max error: 0.00e+00)
Iter 76 [ 10x 38, density=0.11, nnz=  41]: PASS (Max error: 0.00e+00)
Iter 77 [ 11x 29, density=0.07, nnz=  20]: PASS (Max error: 0.00e+00)
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Iter 77 [ 11x 29, density=0.07, nnz=  20]: PASS (Max error: 0.00e+00)
Iter 78 [ 26x 43, density=0.18, nnz= 233]: PASS (Max error: 0.00e+00)
Iter 79 [  7x 24, density=0.10, nnz=  23]: PASS (Max error: 0.00e+00)
Iter 80 [ 42x 18, density=0.31, nnz= 231]: PASS (Max error: 0.00e+00)
Iter 81 [ 28x 14, density=0.39, nnz= 149]: PASS (Max error: 0.00e+00)
Iter 82 [ 14x 34, density=0.28, nnz= 141]: PASS (Max error: 0.00e+00)
Iter 83 [ 35x 36, density=0.27, nnz= 339]: PASS (Max error: 0.00e+00)
Iter 84 [ 10x 45, density=0.06, nnz=  15]: PASS (Max error: 0.00e+00)
Iter 85 [ 10x 15, density=0.12, nnz=  17]: PASS (Max error: 0.00e+00)
Iter 86 [ 30x  6, density=0.39, nnz=  67]: PASS (Max error: 0.00e+00)
Iter 87 [ 37x 21, density=0.24, nnz= 168]: PASS (Max error: 0.00e+00)
Iter 88 [ 36x 43, density=0.06, nnz=  98]: PASS (Max error: 0.00e+00)
Iter 89 [ 16x 12, density=0.08, nnz=  15]: PASS (Max error: 0.00e+00)
Iter 90 [ 19x  6, density=0.09, nnz=  10]: PASS (Max error: 0.00e+00)
Iter 91 [ 13x 11, density=0.09, nnz=  19]: PASS (Max error: 0.00e+00)
Iter 92 [ 32x 36, density=0.16, nnz= 188]: PASS (Max error: 0.00e+00)
Iter 93 [  9x 37, density=0.39, nnz= 155]: PASS (Max error: 0.00e+00)
Iter 94 [ 23x 15, density=0.20, nnz=  72]: PASS (Max error: 0.00e+00)
Iter 95 [  8x 21, density=0.13, nnz=  25]: PASS (Max error: 0.00e+00)
Iter 96 [ 16x 22, density=0.17, nnz=  52]: PASS (Max error: 0.00e+00)
Iter 97 [ 39x 32, density=0.37, nnz= 467]: PASS (Max error: 0.00e+00)
Iter 98 [  9x 20, density=0.14, nnz=  28]: PASS (Max error: 0.00e+00)
Iter 99 [ 10x 13, density=0.30, nnz=  39]: PASS (Max error: 0.00e+00)

All tests passed! (100/100 iterations passed)

~/work via C v13.3.0-gcc
```